



allegro

Pixels, Vectors, or Text? **Allegro Search Engine Goes Multimodal**

Marcin Wiktorowski, ML Research Engineer

Allegro Tech Talks | AI Edition 17.03.2026 | Poznań

allegro

Rozwiń talent

z e-Xperience!



E-XPERIENCE



6 miesięcy
praktycznych zadań
(lipiec-grudzień)



Umowa o pracę



Wsparcie buddy'ego

**Rekrutacja
wystartowała! Aplikuj
na jobs.allegro.eu**

Bądź na bieżąco, zaobserwuj nas na social media



INSTAGRAM



FACEBOOK

@allegrodobrzetubyc @Allegro Tech
@Allegro Jobs

Kogo poszukujemy w programie e-Xperience?



→ **Junior Automation Engineer**

→ **Junior Data Engineer/Data Scientist**

→ **Junior Software Engineer (Java/Kotlin, Python, Mobile)**

Pełna lista stanowisk wraz ze
szczegółami na Allegro Jobs

Polecamy zeskanować kod QR 🔥



AI HUB

- **70+ experts**
(Engineering, Data,
Research & Product)
- **Impactful AI
applications &
cutting edge
research**
(ml.allegro.tech)

Product Analyst

Poznań, Nowy Rynek

Data Scientist – Central Query Understanding

Poznań, Nowy Rynek

Senior Research Engineer – Learning to Rank

Warszawa, Fabryka Norblina

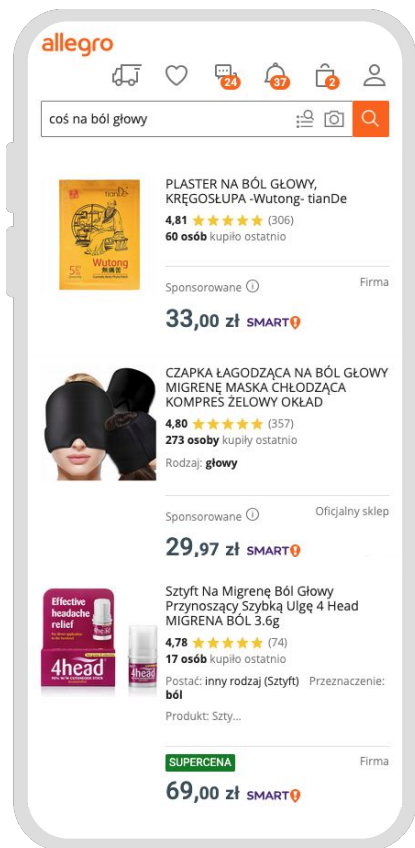
Senior Software Engineer (Java / Kotlin)

Poznań, Nowy Rynek

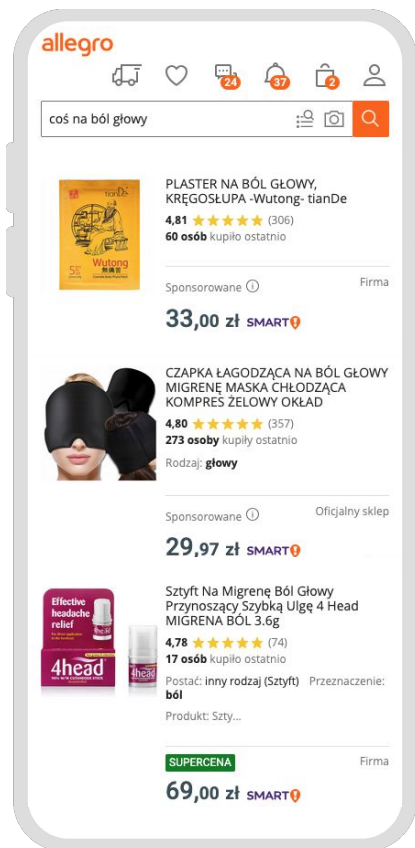


allegro

Lexical search



Lexical search



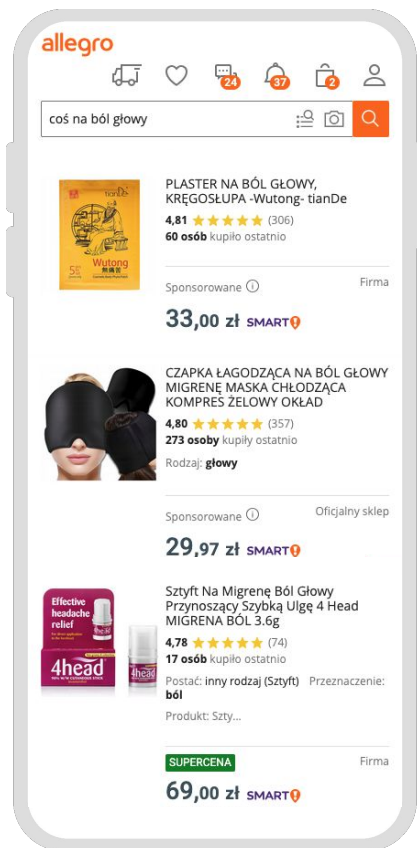
something for a headache

headache patch

cap for headaches

headache relief stick

Lexical search



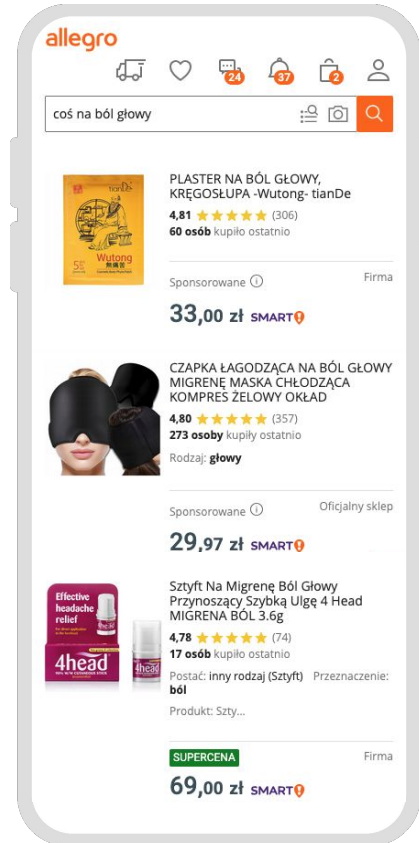
something for a **headache**

headache patch

cap for **headache**

headache relief stick

Lexical search

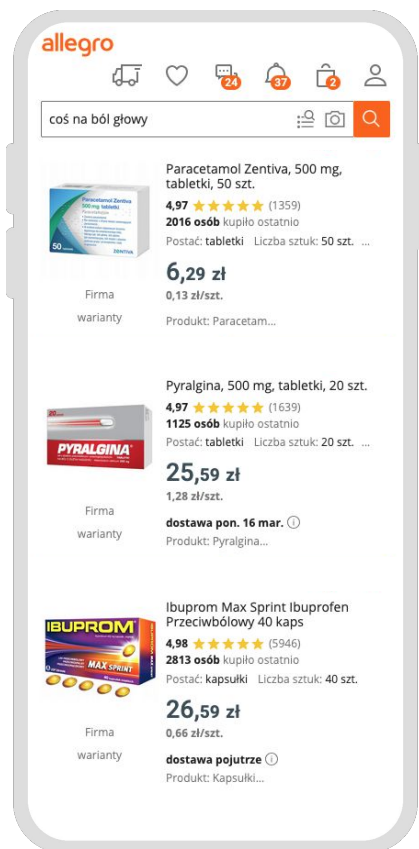


What's the problem solved by lexical search?

- **Directly matches the words** in the search query with the words in the product title
- Only find products containing the word 'headache'
- Does not understand synonyms, context or intent

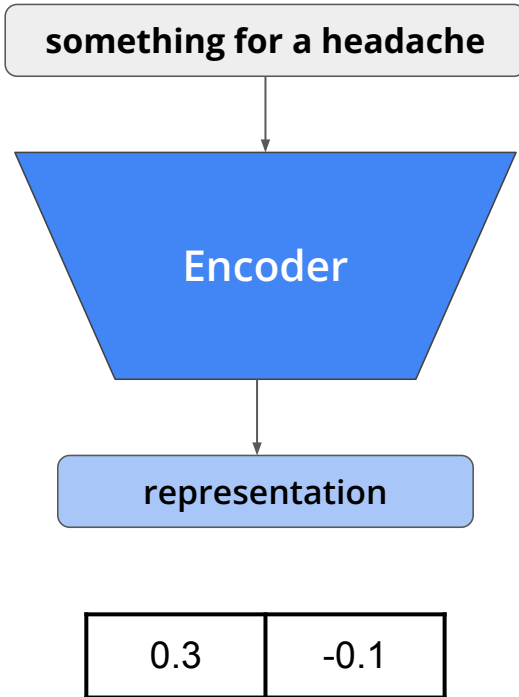


Semantic search

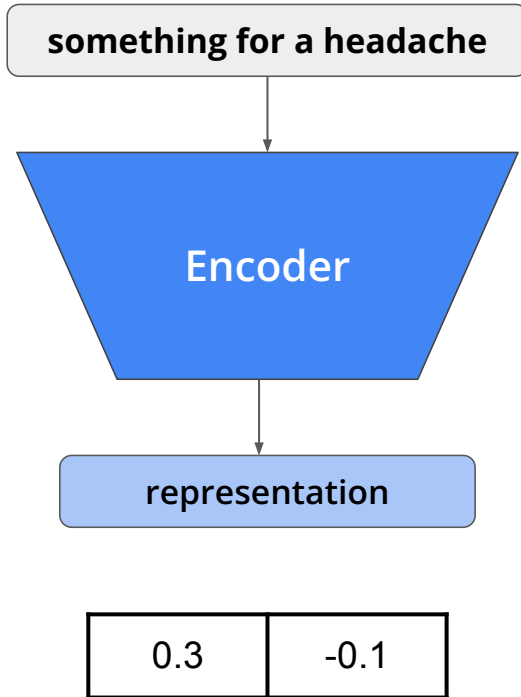


- **Understands the semantic meaning** of the query
- Will also find “paracetamol”, “ibuprofen” and “pyralgin”
- Operates on a mathematical representation of language

Key idea: convert text into vector

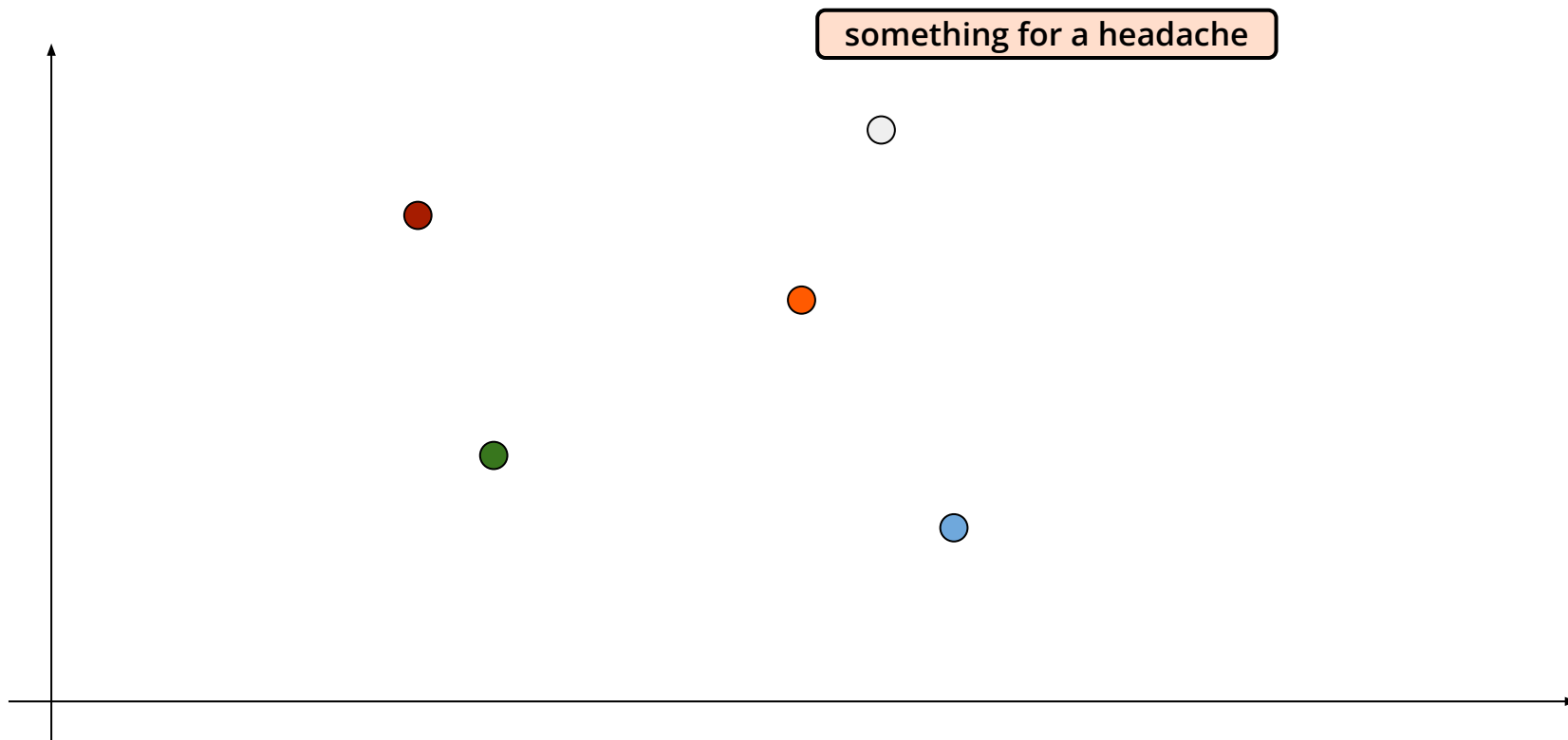


Key idea: convert text into vector

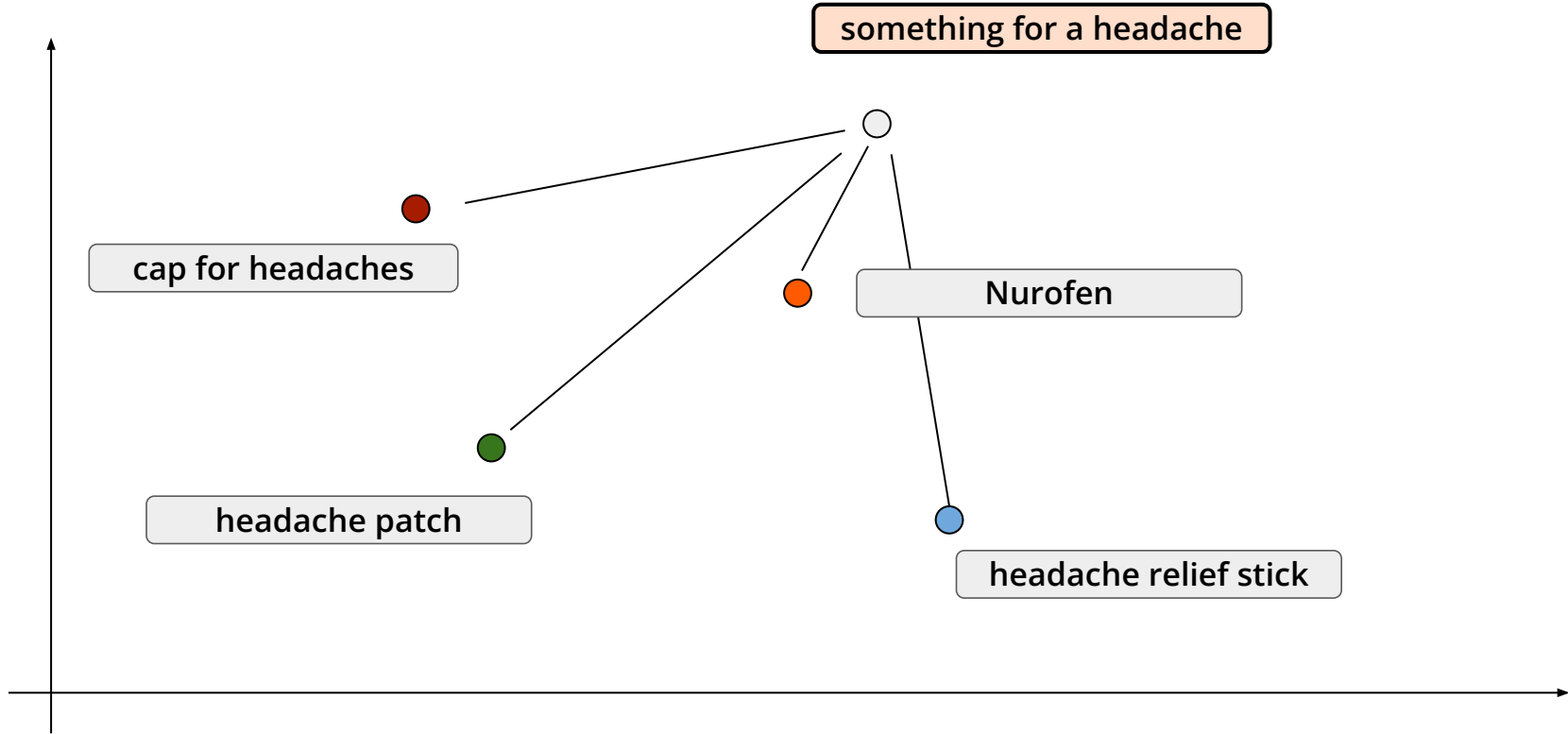


something for a headache	0.3	-0.1
headache patch	0.1	0.1
cap for headaches	0.9	0.2
headache relief stick	-0.4	0.6
Nurofen	0.4	-0.2

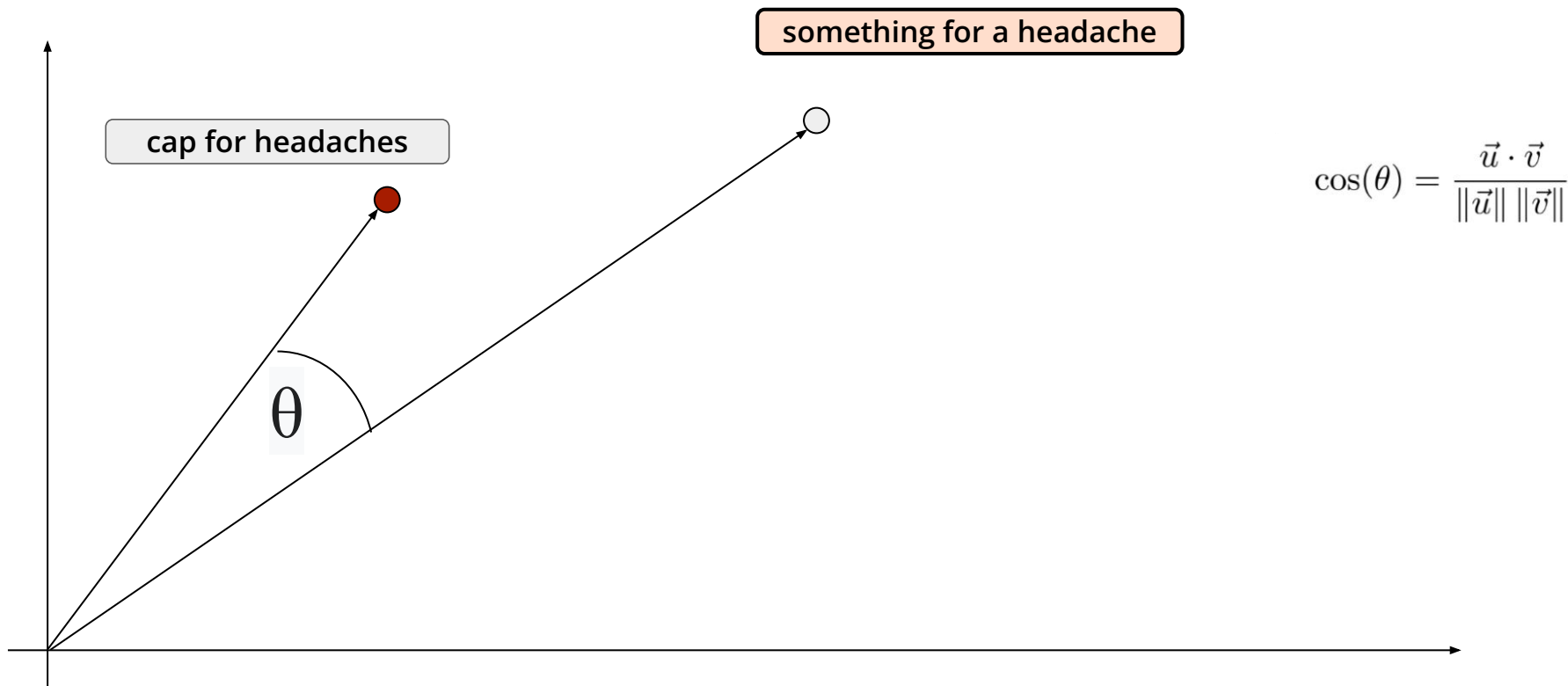
Key idea: convert text into vector



Key idea: convert text into vector



Key idea: convert text into vector



Key idea: convert text into vector

something for a headache

0.3

-0.1

headache patch

0.1

0.1

$$\cos(\theta) = \mathbf{0.44}$$

cap for headaches

0.9

0.2

$$\cos(\theta) = \mathbf{0.87}$$

headache relief stick

-0.4

0.6

$$\cos(\theta) = \mathbf{0.78}$$

Nurofen

0.4

-0.2

$$\cos(\theta) = \mathbf{0.98}$$

Key idea: convert text into vector

something for a headache

1 Nurofen $\cos(\theta) =$ **0.98**

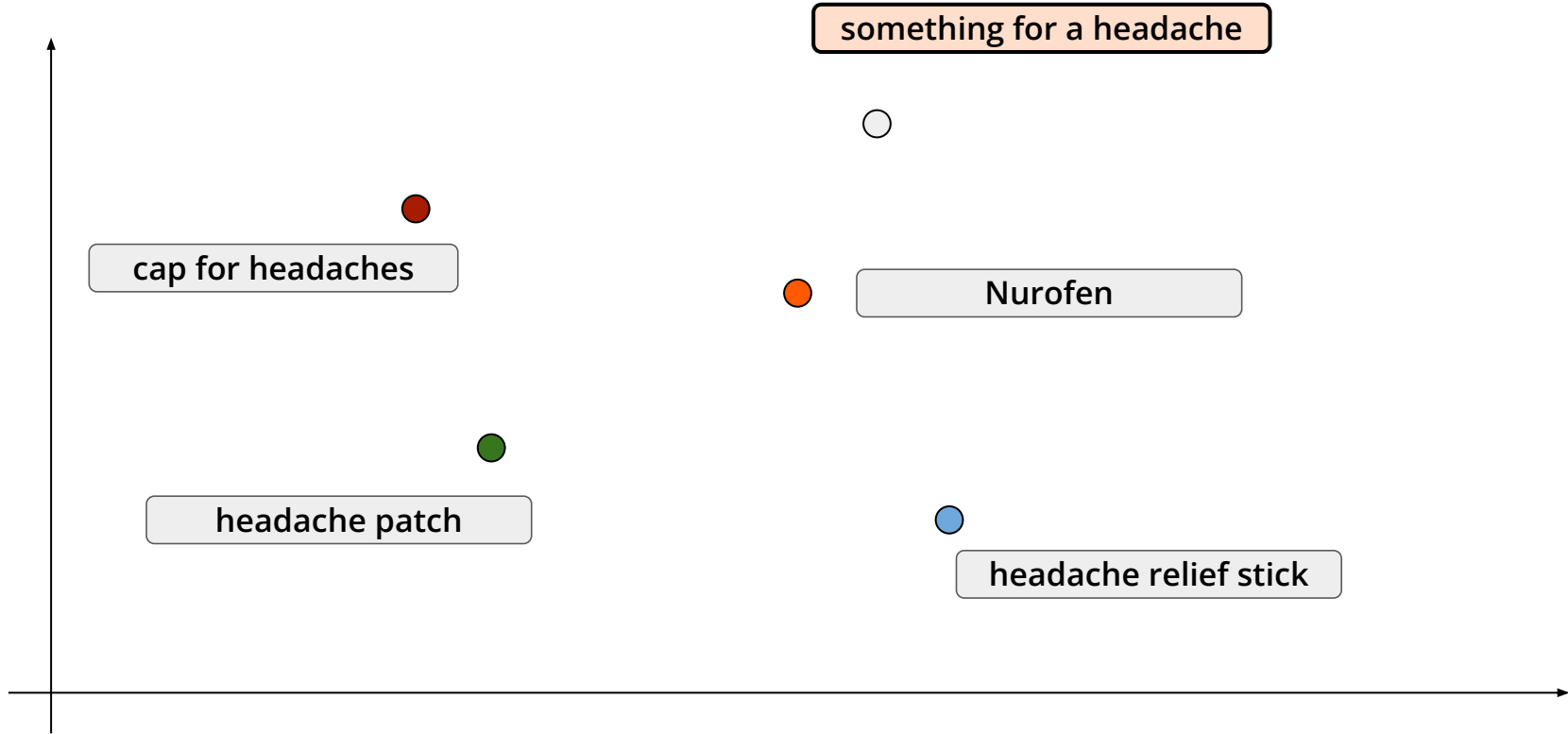
2 cap for headaches $\cos(\theta) =$ **0.87**

3 headache relief stick $\cos(\theta) =$ **0.78**

4 headache patch $\cos(\theta) =$ **0.44**

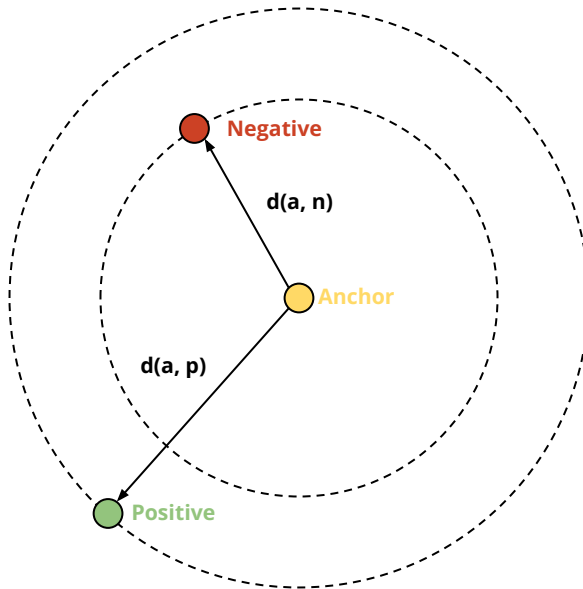


How does the model know what is closest?

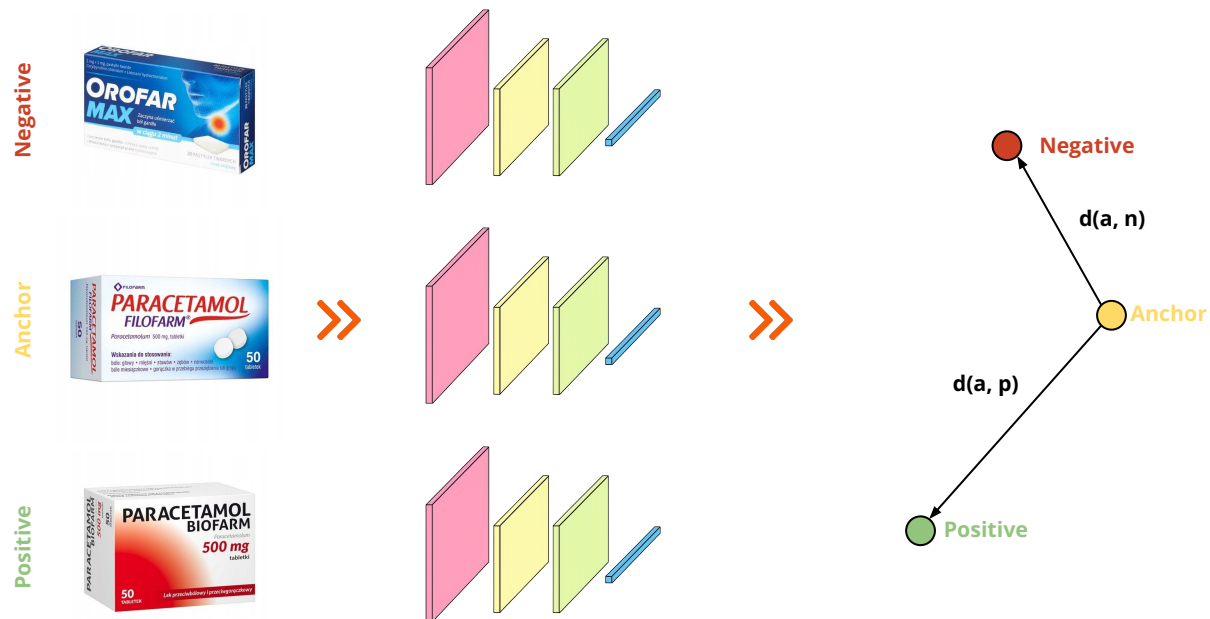


How does the model know what is closest?

Contrastive learning

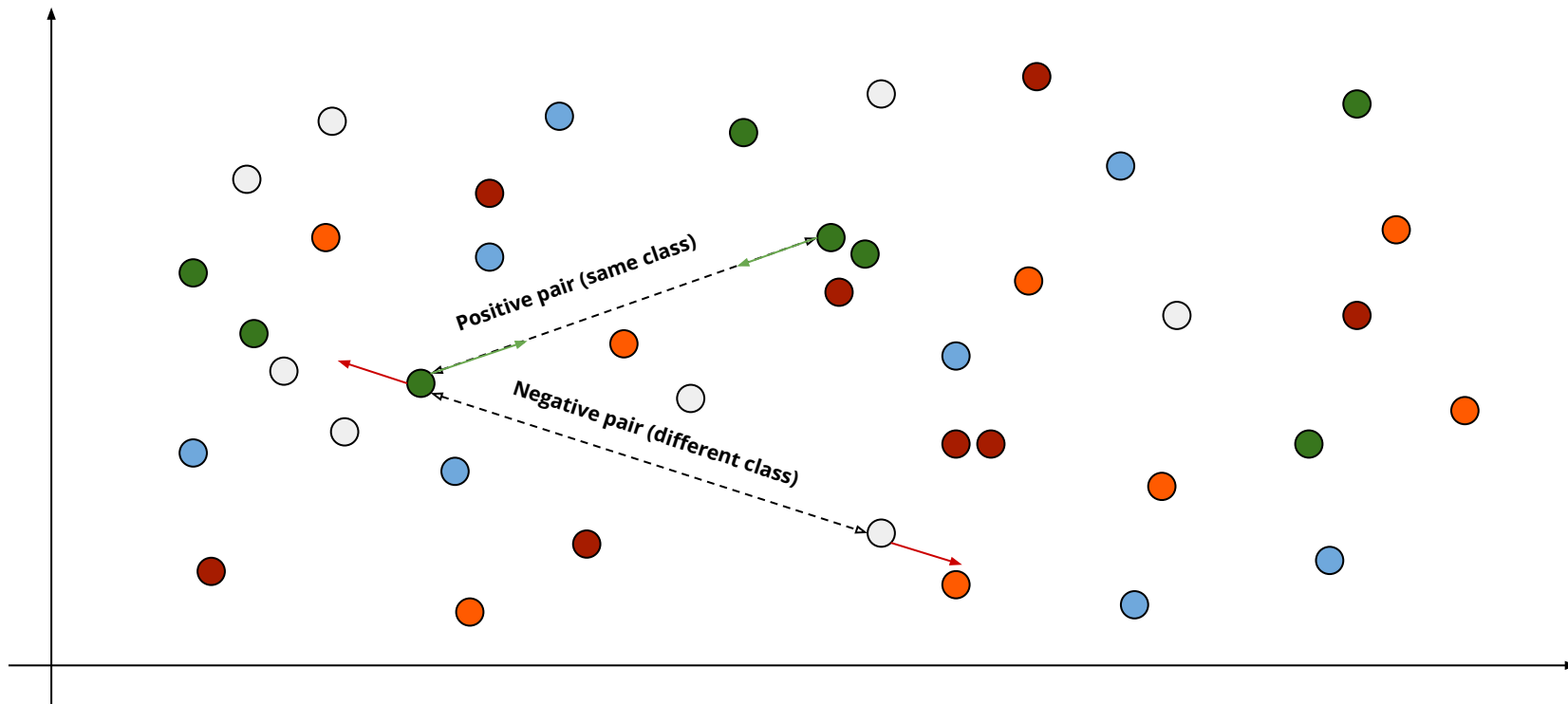


Generating representations



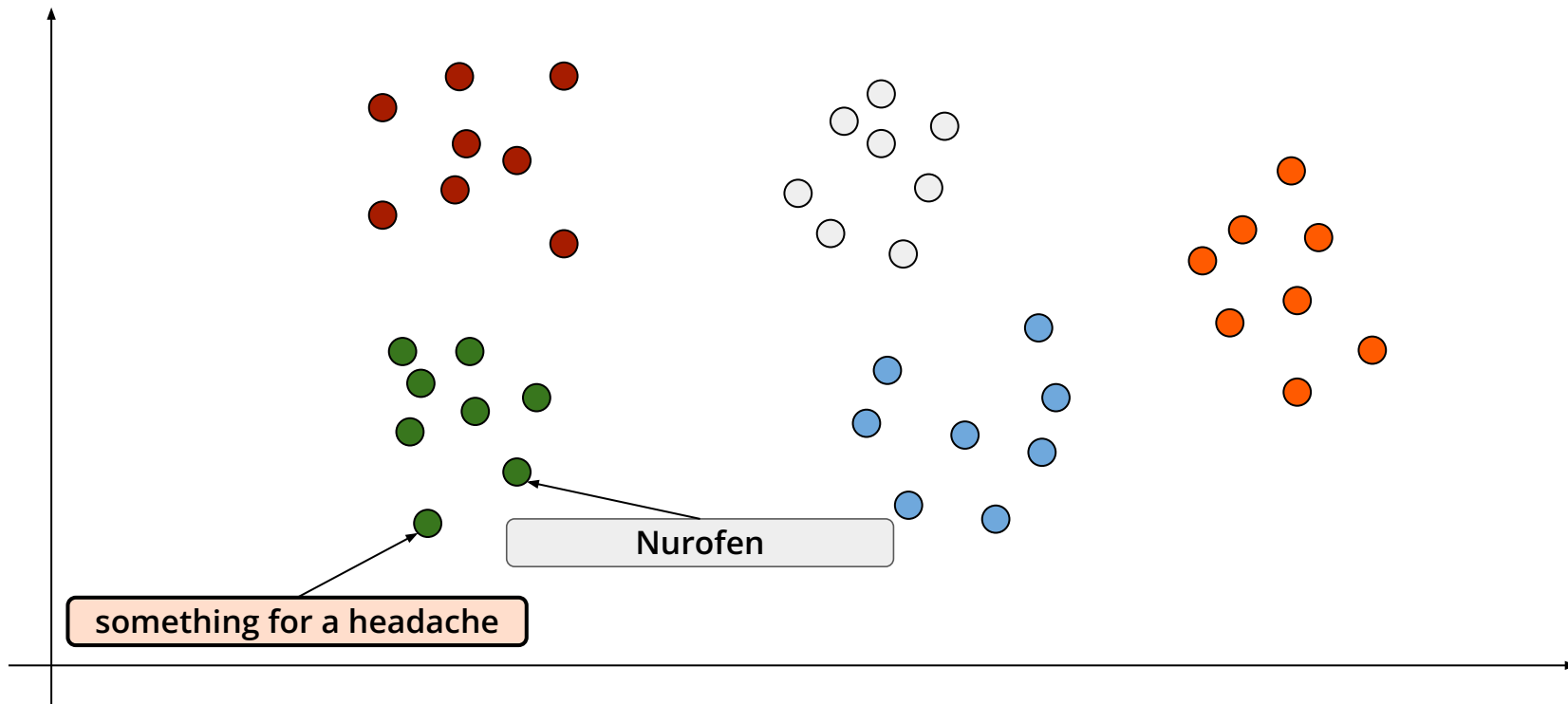
Representation space

Random beginning



Representation space

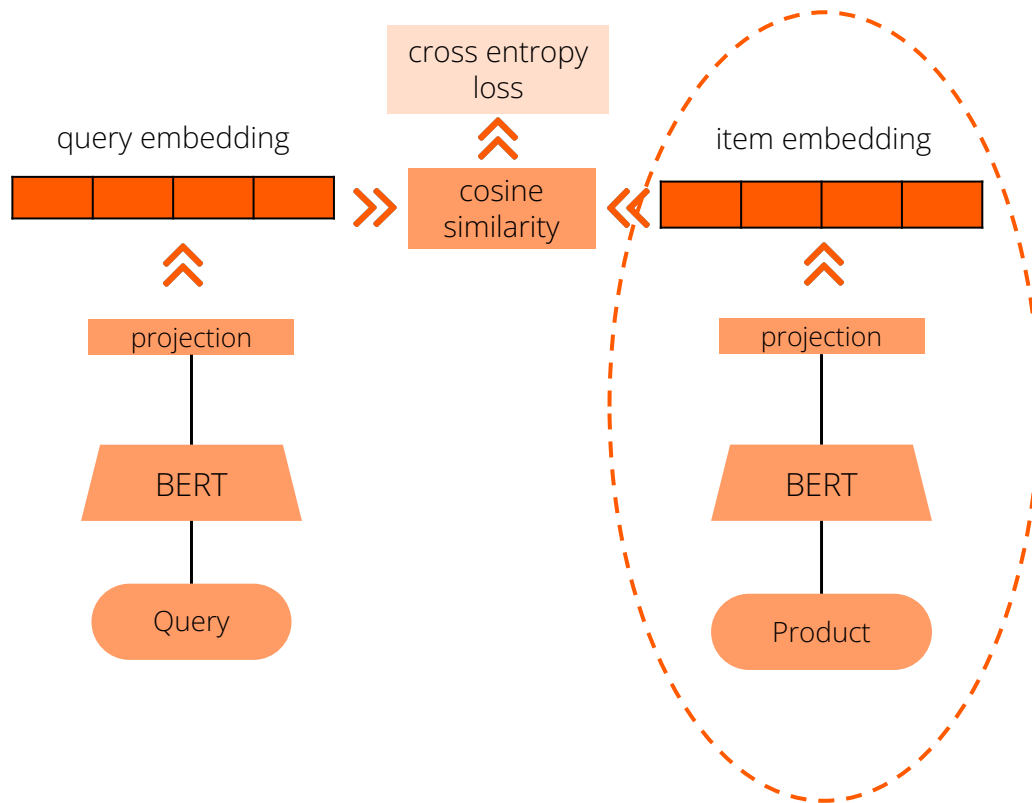
Clustered end



Two-tower model

Contrastive learning inside Allegro semantic-search model

- Query tower:
„winter jacket" → [0.12, -0.45, 0.78, ...]
- Item tower:
"Columbia puffy jacket" → [0.11, -0.42, 0.80, ...]
- **Shared representation space for queries and products**
- A **BERT** text encoder pre-trained on Allegro data
- Why two towers? **Latency is king**
 - item tower computed offline
 - query tower computed online

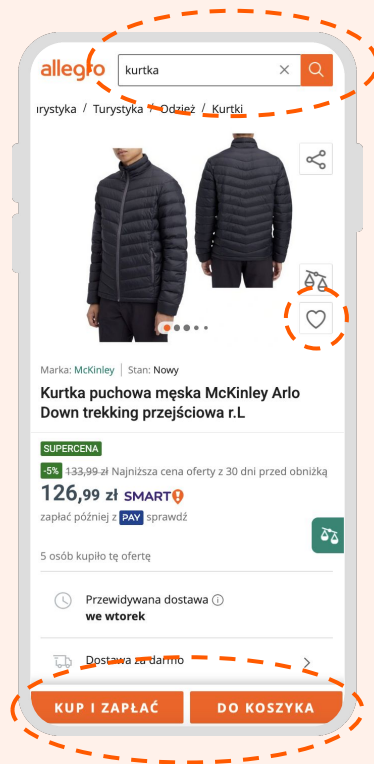


Data

How do we get the data for training?

Challenging **training scale**:

- ~100 million query-product pairs
- Multi-GPU
- Batch size > 70,000 - **most important HP!**
- Several days of training



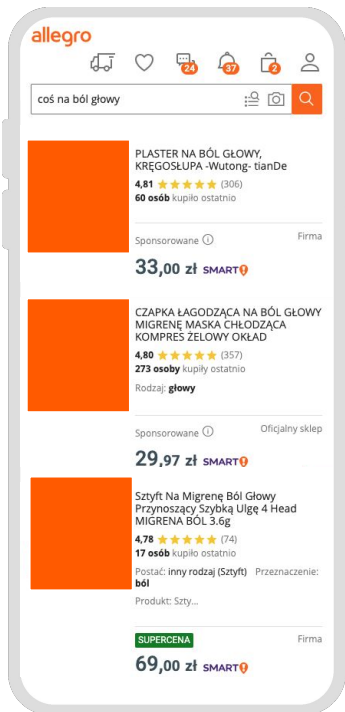
Query / Product pairs via:

- Clicks
- Purchases
- Add to cart
- Add to favourites

Multimodal search? But why?

Text vs image

Text 1:0 Image



Text only

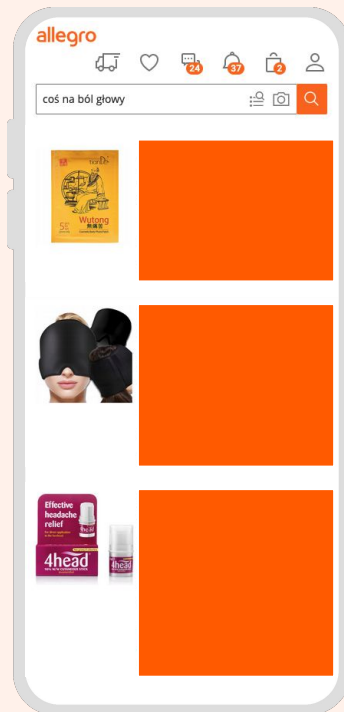
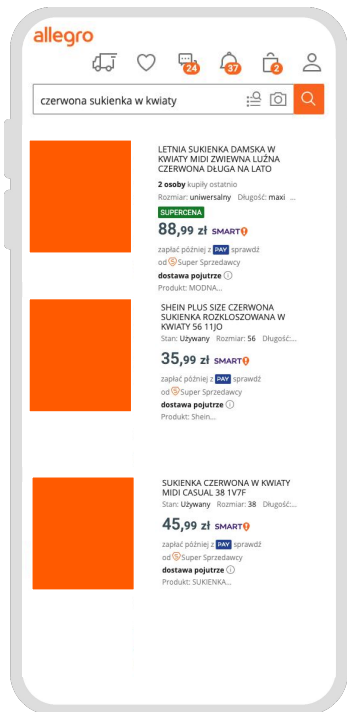


Image only

Text vs image

Text 1:1 Image



Text only

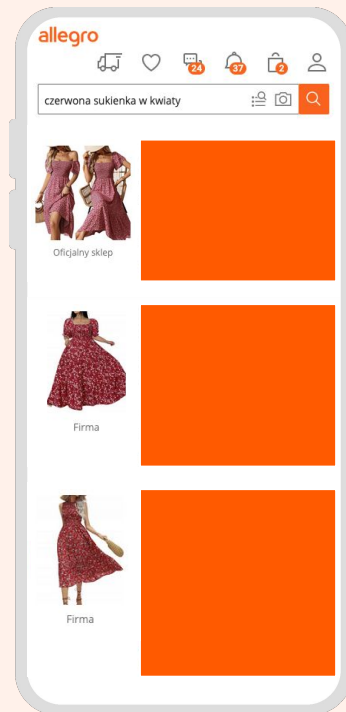


Image only

Images

Why?

- Users ask queries with specific attributes (e.g. color)
- Images contain complementary information not available in text (**details, texture, style** etc.)
- We primarily assess e-commerce search results by images.



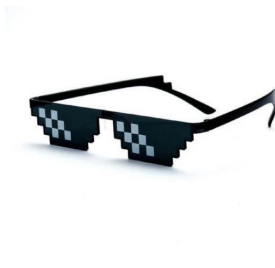
Query: minecraft gadgets

Text-only model



Product name:
Minecraft mug and cushion
set for player for fan gift +
name

Multimodal model



Product name:
Thug Life sunglasses
pixel

Impact assessment

The analogy of drilling into rock



Impact assessment

Estimating the importance of visual modality through manual data annotation



Query: klocek brown lego otwór

Query category: None

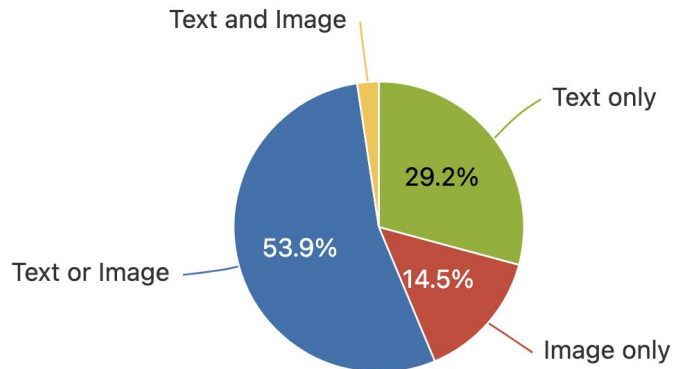
Product name: Lego Technic 3700 Belka 1X2 Brązowa (1G)

Product category: Elementy

Product image captions: brązowy klocek lego 1x2 z otworem na oś

☐ text-or-image^[1] ☐ text-and-image^[2] ☐ text-only^[3] ☐ image-only^[4] ☐ neither^[5]

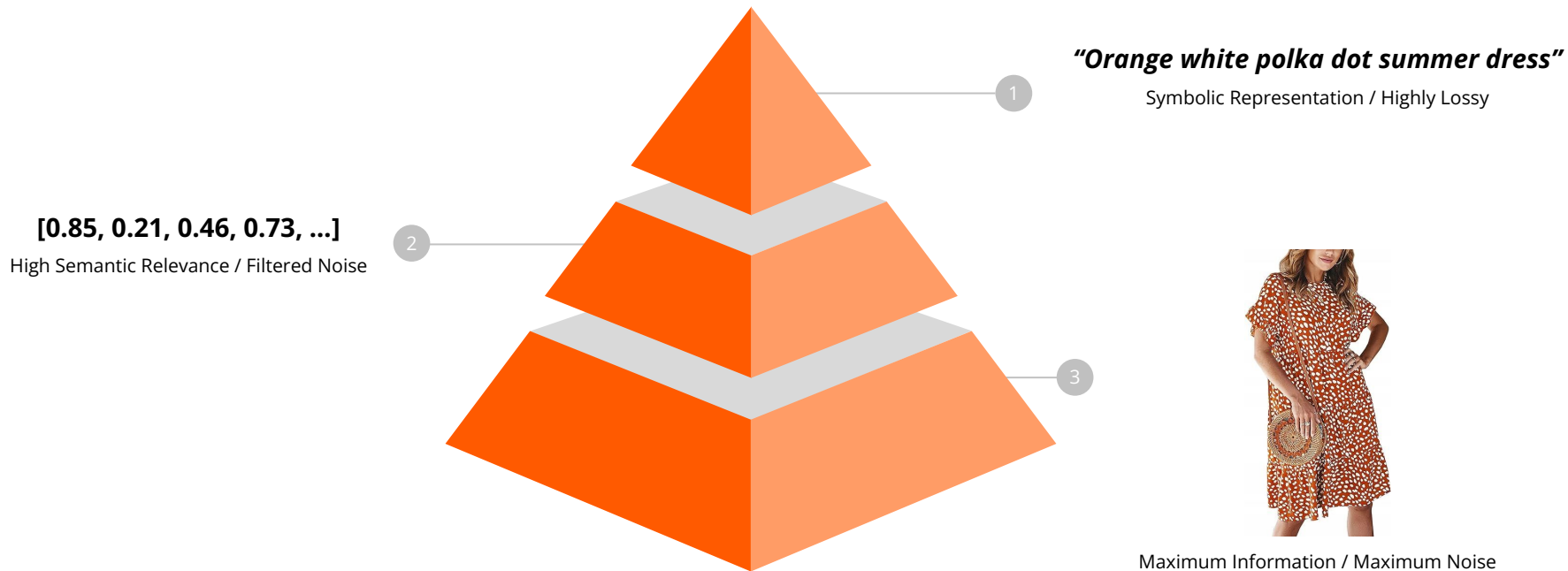
Class distribution without "neither" class



Insight: visual modality is **required for ~17%** of relevant query-product pairs.

The information "richness" trade-off

How much reality can we afford?



Literature

How do industry/academia do it?

End-to-end multi-modal product matching in fashion e-commerce

Captions Speak Louder than Images: Generalizing Foundation Models for E-commerce from High-quality Multimodal Instruction Data

Multimodal semantic retrieval for product search

MRSE: An Efficient Multi-modality Retrieval System for Large Scale E-commerce

Hao Jiang^{1,3}, Haoxiang Zhang³, Qingshan Hou², Chaofeng Chen¹,
Weisi Lin^{1*}, Jingchang Zhang³, Annan Wang¹

¹Nanyang Technological University (NTU), Singapore

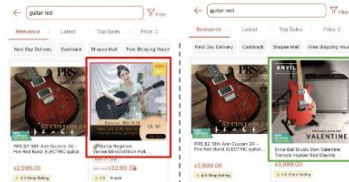
²National University of Singapore (NUS), Singapore

³Shopee, Singapore

{jianghao907, haoxiang.z.f, houqingshanv, chaofenghust}@gmail.com,

Abstract

Providing high-quality item recall for text queries is crucial for developing search systems in large-scale e-commerce. Current solutions employ Embedding-based Retrieval Systems (ERS) to embed queries and items into a low-dimensional space. However, uni-modality ERS is overly reliant on textual features, rendering it unreliable in complex semantic contexts. Existing multi-modality ERS often neglects individual preferences for different modalities and fails to fully leverage their advantages. To address these



MAKE: Vision-Language Pre-training based Product Retrieval in Taobao Search

Generalized Contrastive Learning for Multi-Modal Retrieval and Ranking

Captions Are Worth a Thousand Words: Enhancing Product Retrieval with Pretrained Image-to-Text Models

CommerceMM: Large-Scale Commerce MultiModal Representation Learning with Omni Retrieval

Licheng Yu, Jun Chen, Animesh Sinha, Mengjiao MJ Wang, Yu Chen, Tamara L. Berg, Ning Zhang

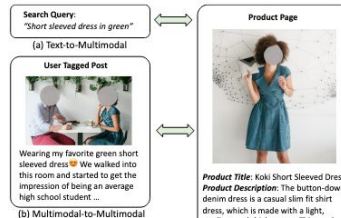
Facebook AI

Menlo Park, California, USA

{lichengyu,junchen20,animeshsinha,mengjiaow,hugochen,tlberg,ningzhang}@fb.com

ABSTRACT

We introduce CommerceMM - a multimodal model capable of providing a diverse and granular understanding of commerce topics associated to the given piece of content (image, text, image+text), and having the capability to generalize to a wide range of tasks, including Multimodal Categorization, Image-Text Retrieval, Query-to-Product Retrieval, Image-to-Product Retrieval, etc. We follow the pre-training + fine-tuning training regime and present 5 effective pre-training tasks on image-text pairs. To embrace more common and diverse commerce data with text-to-multimodal, image-to-multimodal, and multimodal-to-multimodal mapping, we propose another 9 novel cross-modal and cross-pair retrieval tasks, called Omni Retrieval, and training. The pre-training is conducted in an



...and a lot more knowledge - let's try to organise it

Literature review

Let's try to organise the knowledge

- **~30 publications on multimodal retrieval**, both in and outside of e-commerce
- Automatic synthesis using MLLM (Gemini Pro 2.5):

Please answer the following questions:

- What are the affiliations of the authors (e.g., Alibaba, Amazon, Meta, Academia)?
- What is the year of publication?
- Does the paper address a retrieval task? If so, is it specifically for e-commerce?
- What are the input modalities for the query and the item?
- What encoders are used in the best-performing variants of the model?
- Is the vision encoder fine-tuned before final training?
- What are the objectives for pre-training, fine-tuning, and the final training?
- During training, which parts of the model are trained, and which parts are frozen?
- What is the embedding dimension of the visual encoder?
- Does the visual encoder have a projection layer before the fusion step?
- What type of fusion method is used?
- Does the model use a single [CLS] token/projection from the visual encoder, or does it use the entire sequence of tokens?
- What was the training infrastructure (e.g., GPUs), and what was the batch size?
- What are some other interesting or novel aspects of this paper?

Here's the paper:

Literature review

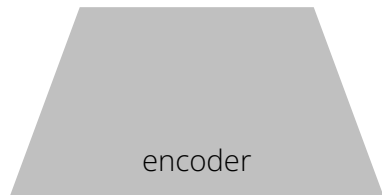
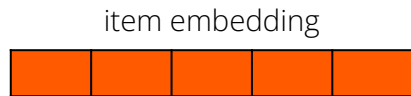
Our findings

1. **End2end approaches usually outperform** approaches with static embeddings and image captions.
2. All three approaches (end2end, static embeddings, image captions) are used in industry. **End2end training is still dominating.**
3. The larger the company, the higher probability the **training will be end2end.**
4. But end2end training usually **requires wild infrastructure** (dozens of GPUs).

Paper	Year	Author	Modalities
Retrieval inside e-commerce			
UniECS: Unified Multimodal E-Commerce Search Framework with Gated Cross-modal Fusion 🍷🌟	2025	Kuaishou	Query: Text, Image Item: Text, Image
MOON: Generative MLLM-based Multimodal Representation Learning for E-commerce Product Understanding 🍷	2025	Alibaba	Query: Text or Image Item: Text or Image
Multimodal semantic retrieval for product search	2025	Amazon	Query: Text Item: Text, Image
VL-CLIP: Enhancing Multimodal Recommendations via Visual Grounding and LLM-Augmented CLIP Embeddings	2025	Walmart	Query: Text Item: Image
Bringing Multimodality to Amazon Visual Search System	2024	Amazon	Query: Text, Image Item: Text, Image

Business context: take into consideration training time, latency overhead, implementation cost, and maintenance aspects (refreshing the model). The simpler the better.

Semantic search goes multimodal



Item Tower



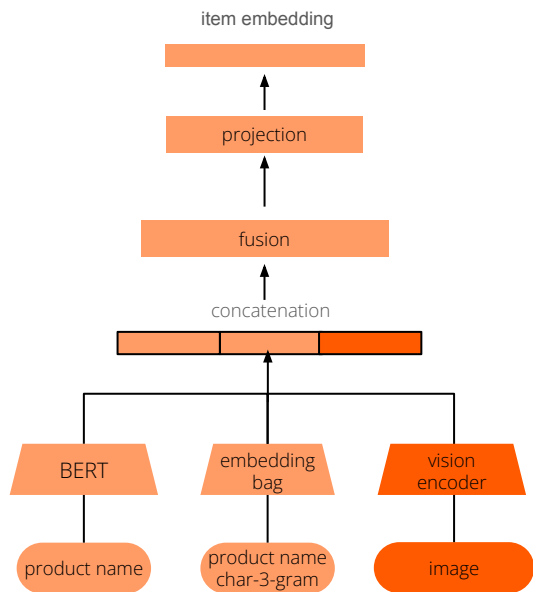
Set of Pikachu
Pokémon Go Balloons
Large 18 pcs
50cm x 65cm



Formally:

$$\mathbf{h}_{item} = f_{text}(x_{text}) \gg \mathbf{h}_{item} = \mathcal{F}_{fusion}\left(f_{text}(x_{text}), f_{vision}(x_{vision})\right)$$

Images - end2end training

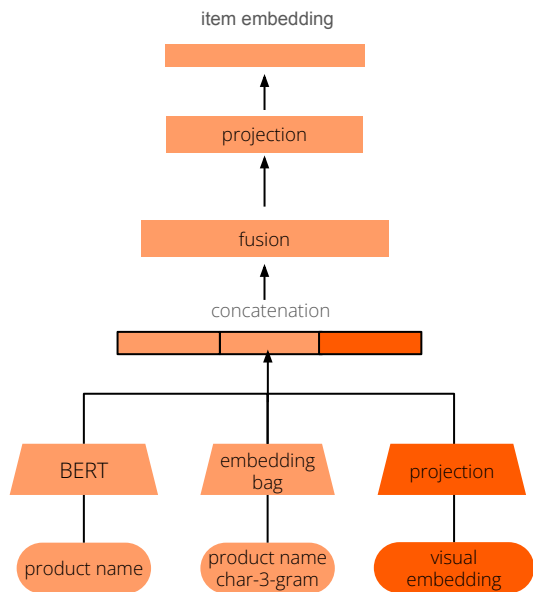


dynamic = trainable

How?

- **Embeddings** produced by a trainable vision encoder
- Initialize visual backbone with open-source or in-house models
- Most powerful setup to learn from data
- Vision models - **high GPU usage** during training - **lower batch size** (trade-off in favour of text-only model)

Images - static embeddings

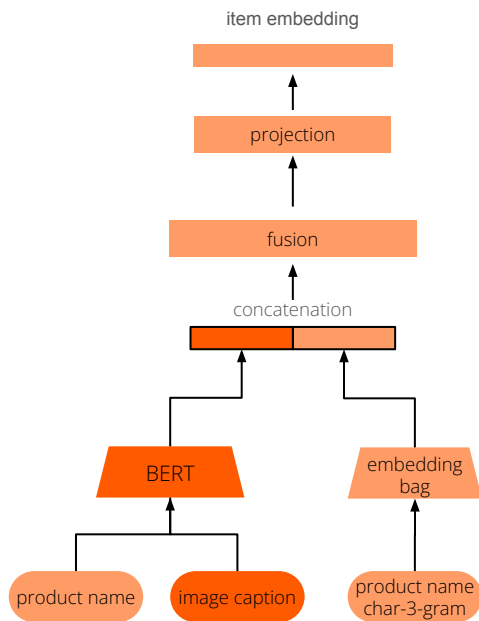


static = non trainable

How?

- **Static visual embeddings**
- **Reasonable GPU usage** during training by pre-computing embeddings (better trade-off)
- Problem with **modality utilization** - limited adaptability

Images - image captions



How?

- Generate **image captions** via **VLM**
- **Bypass modality utilization** problem - simply add image caption to product name (BERT input)
- Possibility to **lose some visual information**, bias of prompt engineering and model choice.

The Engineering Reality

Iteration #1

End2end vs static embeddings

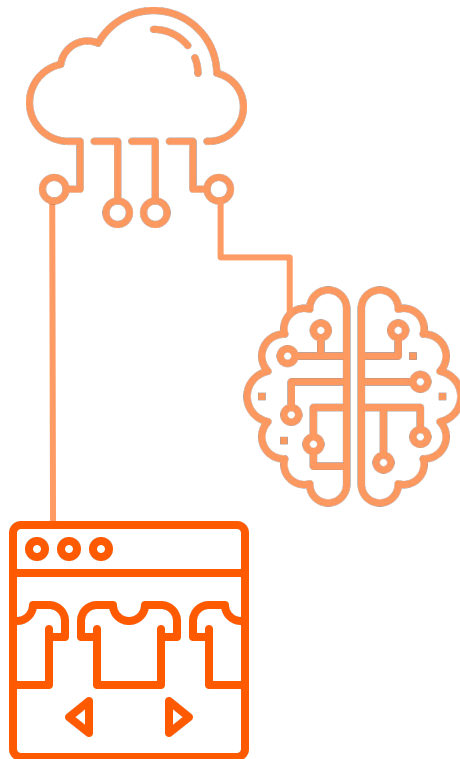
Visual encoders:

- allegro/visual-search-efficientnet
- openai/clip-vit-large-patch14
- mistralai/pixtral-12b-vit
- marqo/fashion-siglip-vit-base-patch16
- google/siglip-so400m-patch16 🏆

End2end: immediate failure ❌ due to batch size explosion



Static embeddings: partial success ✅ but requires optimization



Iteration #1.5

Image captions

- A decent improvement, **but...**
- Data generation problems for several million products (model availability, cost)
- Context window explosion 💣 - increased memory requirements
- Larger batch size >> image captions

Verdict: ✅ but an iceberg of problems 🧊

skrzynce, z pomarańczowymi
dźwigniami, do łączenia przewodów
elektrycznych.



Czarna lampa wisząca w stylu
industrialnym z elementami w
kolorze miedzi, z regulacją
wysokości za pomocą systemu
bloczków.



Zestaw dwóch czarnych tonerów 4ink
Premium Toner Q2612A,
kompatybilnych z drukarkami
laserowymi, z 10-letnią gwarancją.



Karta kolekcjonerska FIFA 365,
przedstawiająca Manuela Lanzini
w koszulce River Plate z logo 'codere',
w kolorach białym i czerwonym.



Czarna lampa garażowa LED z
regulowanymi panelami, idealna do
oświetlenia warsztatu lub garażu.



na białej szpul, idealna do
pakowania prezentów i dekoracji.



Nawóz YaraMila Complex w białym
opakowaniu z zielonymi elementami
graficznymi i logo Yara.



Apple Watch Ultra 49mm z tytanową i
ceramiczną obudową, szafirowym
kryształem, GPS, LTE, WR-100M, DIVE-
40M i zielonym paskiem.



Modality utilization / imbalance

Chen Henry Wu^{*,1} Neil Kale^{*,1} Aditi Raghunathan¹

1. *Journal of the American Medical Association*, 1997; 278: 1021-1025.

Improving Multimodal Learning via Imbalanced Learning

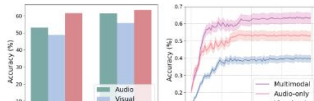
DynCIM: Dynamic Curriculum for Imbalanced Multimodal Learning

Chengxuan Qian¹, Kai Han¹, Jiaxin Liu², Zhenlong Yuan^{3,4}, Zhengzhong Zhu⁵,
Jingchao Wang⁶, Chongwen Lyu¹, Jun Chen¹, Zhe Liu^{1,†}

¹Jiangsu University ²UIUC ³Alibaba ⁴UCAS ⁵Sichuan University ⁶Peking University
chengxuan.qian@stmail.ujs.edu.cn, zliu@ujs.edu.cn

Abstract

Multimodal learning integrates complementary information from diverse modalities to enhance the decision-making process. However, the potential of multimodal collaboration remains under-exploited due to disparities in data



MMPareto: Boosting Multimodal Learning with Innocent Unimodal Assistance

Modality-Balanced Embedding for Video Retrieval

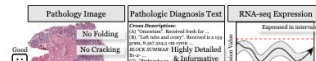
Xun Wang¹ Bingqing Ke¹ Xuanping Li¹ Fangyu Liu²
Mingyu Zhang¹ Xiao Liang¹ Qiushi Xiao¹ Cheng Luo¹ Yue Yu¹
¹ Kuaishou ² University of Cambridge

Modalities Contribute Unequally: Enhancing Medical Multi-modal Learning through Adaptive Modality Token Re-balancing

Jie Peng^{*1} Jenna L. Ballard^{*2} Mohan Zhang³ Sukwon Yun³ Jiayi Xin² Qi Long² Yanyong Zhang¹
Tianlong Chen³

Abstract

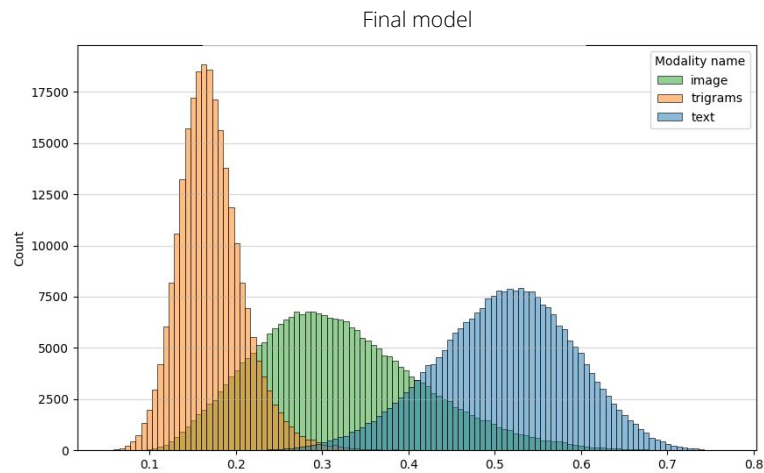
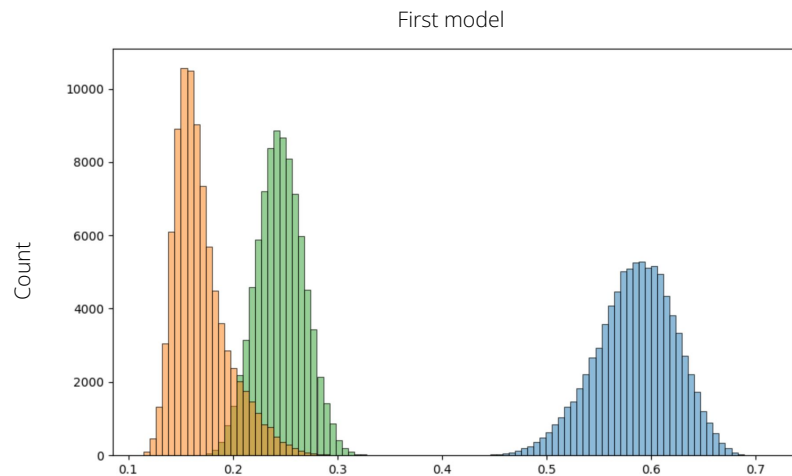
Medical multi-modal learning requires an effective fusion capability of various heterogeneous



Iteration #2

Modality utilization / imbalance

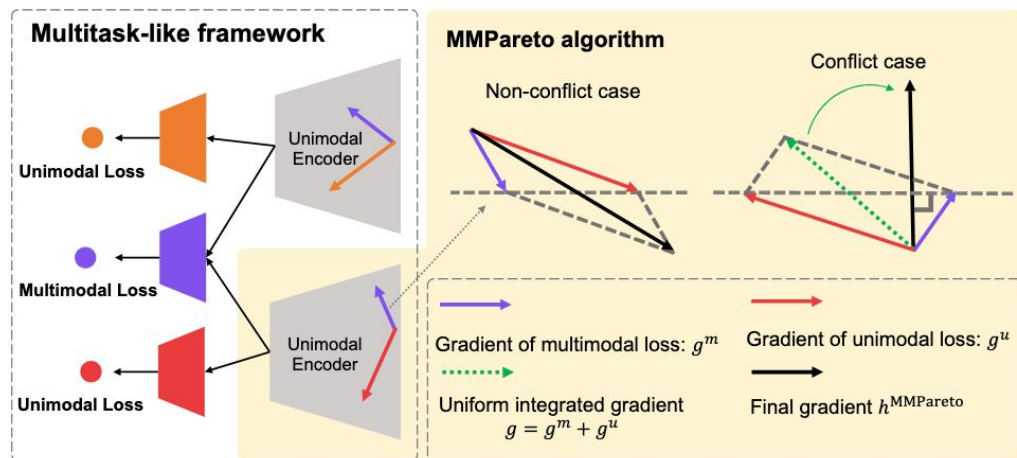
$$[\alpha_{\text{text}}, \alpha_{\text{image}}] = \text{Softmax}\left(W_{\text{proj}}[E_{\text{text}}(x_{\text{text}}) \parallel E_{\text{vision}}(x_{\text{image}})]\right)$$



Attention weights per modality

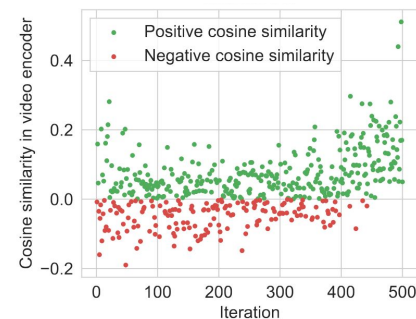
Iteration #2

Modality utilization / imbalance

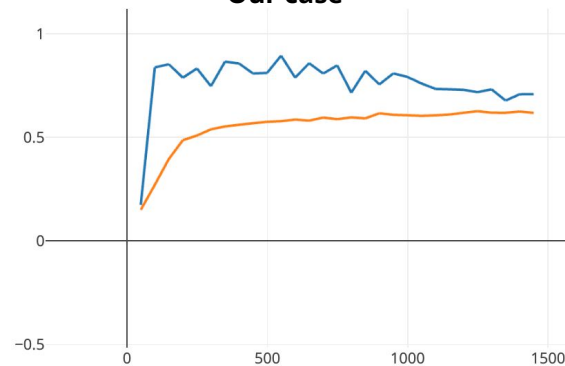


"MMPareto: Boosting Multimodal Learning with Innocent Unimodal Assistance"

Author's case



Our case



Iteration #2

Evaluation / Logging Policy Trap

Context $\mathbf{x}$ = search query

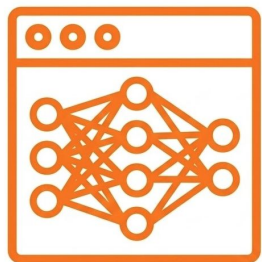
Action $\mathbf{y} = \pi(\mathbf{x})$ = retrieved products

Feedback $\delta(\mathbf{x}, \mathbf{y})$ = click / purchase

Utility function $U(\pi) = ???$



Query: **orange dress**

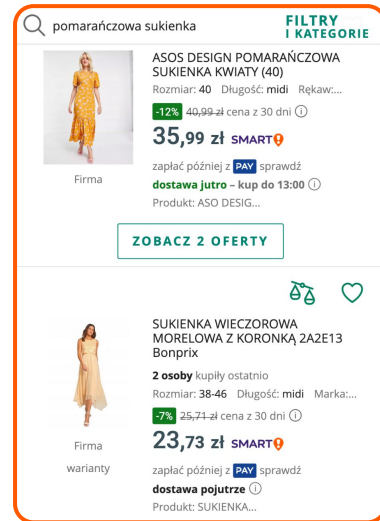


Logging policy π

Context $\mathbf{x}$

Feedback $\delta(\mathbf{x}, \mathbf{y})$

Action $\mathbf{y} = \pi(\mathbf{x})$



previous production policies (text-only search engine)

Iteration #2

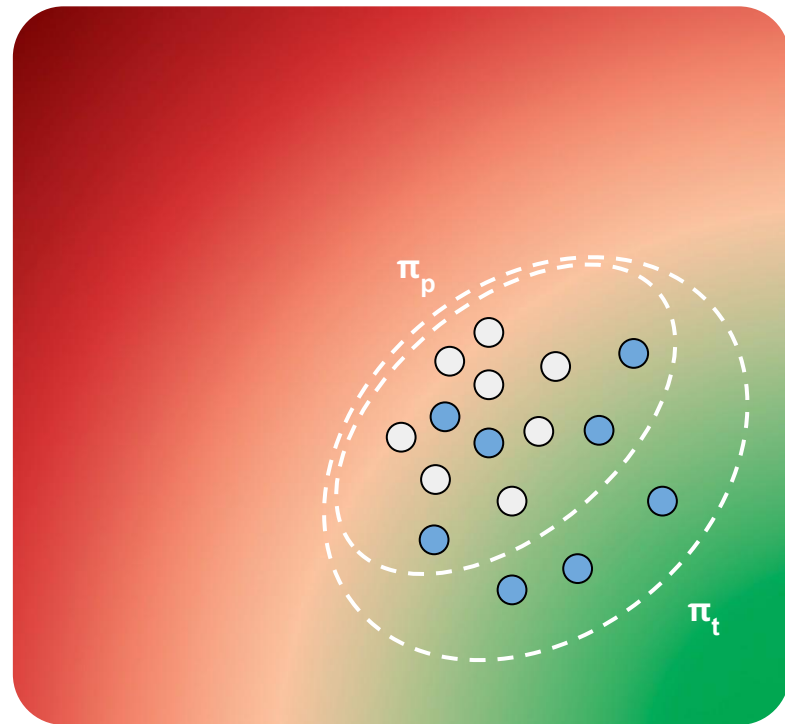
Evaluation / Logging Policy Trap

Problem:

- Not a fully supervised learning task - **clickstream data is inherently noisy**
- Evaluation data is **biased by historical production policies** (text-only search engine)

Solution:

- **A/B tests**
- Offline Policy Evaluation
- Synthetic data!
- Manual annotation



Summary



End2end



Static embeddings

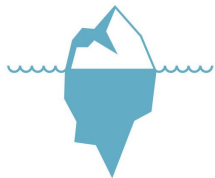


Image captions



Thank you

Questions?

